

The Genotype-Only Ceiling for Factor VIII Inhibitor Risk Prediction: Patient-Level Modelling and Cross-Registry Validation in Haemophilia A

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Abstract—Neutralising alloantibodies ("inhibitors") against infused factor VIII (FVIII) develop in a substantial minority of people with haemophilia A and are the most consequential complication of replacement therapy, yet no risk-stratification tool is in routine pre-treatment clinical use. Machine-learning studies on public *F8* variant registries report accuracies of 97% and above. We show that such figures are not predictive performance, establish what is genuinely achievable, and determine whether the limiting factor is the model or the data. Using the EAHAD/HADB *F8* resource, whose allele-report table contains one row per patient ($n = 10,064$) rather than one row per variant, we assembled 4,966 records carrying a recorded inhibitor outcome across 2,643 variants and 342 reporting studies. We engineered 95 mechanistic descriptors in six blocks and screened twelve model families. All partitions were grouped by variant, no resampling was applied at any stage, and unrecorded outcomes were retained as unlabelled rather than treated as negatives. A rank-averaged ensemble achieved AUC-ROC 0.786 ± 0.004 under repeated variant-grouped cross-validation. The genomic-only rung of the feature ablation reached 0.742, independently reproducing a ceiling previously established on the CDC CHAMP registry; patient baseline phenotype contributed +0.023 AUC. Permuted labels gave 0.504, ungrouped cross-validation inflated the score by 0.032, and a single outcome-derived column from a widely redistributed derived file moved AUC from 0.777 to 0.966. Cross-registry transfer initially appeared to exceed within-registry performance; we traced this to 64.5% variant overlap between CHAMP and EAHAD, and on genuinely novel variants obtained AUC 0.851 (95% CI 0.819–0.881) against 0.624 for CHAMP's own cross-validation on identical rows. Genotype-only prediction saturates near AUC 0.74 in two independently curated registries; because eleven model families span only 0.027 AUC, this is a property of the available data rather than of model capacity.

Index Terms—haemophilia A, factor VIII inhibitors, *F8*, clinical risk prediction, data leakage, external validation, probability calibration.

I. INTRODUCTION

Haemophilia A is an X-linked bleeding disorder caused by variants in *F8*, managed by replacement of the deficient coagulation factor VIII. In a substantial minority of patients the immune system recognises infused FVIII as foreign and produces neutralising alloantibodies, termed inhibitors. Once inhibitors develop, replacement therapy loses efficacy, bleeding becomes markedly harder to control, and the patient requires immune tolerance induction — a prolonged and expensive protocol that does not always succeed.

Inhibitor development is concentrated in the first fifty exposure days. This makes it, in principle, an ideal target for pre-treatment risk stratification: the window in which the outcome occurs is known and short, and the clinical response to elevated risk is available and low-cost. Despite this, no genotype-based stratification tool is in routine clinical use.

The mechanistic basis is well established. Variant classes that abolish endogenous FVIII protein — large deletions, nonsense variants and frameshifts — carry markedly higher inhibitor incidence than missense variants, which permit expression of a full-length, if dysfunctional, protein [14]. The immunological interpretation is that a patient who has never expressed FVIII has established no central tolerance to it, so infused factor is encountered as a wholly foreign antigen.

Several groups have applied machine learning to public *F8* registries, most prominently the CDC Hemophilia A Mutation Project (CHAMP) list [2]. Reported accuracies reach 97.37% [1]. If reproducible as *predictive* performance, such figures would be clinically decisive. They are not reproducible. Prior

work by this group established that these estimates arise from three specific preprocessing choices, and placed the honest ceiling of genotype-only data near AUC 0.74–0.75. That finding raises the question this paper addresses: **is that ceiling a limitation of the modelling, or of the data?**

CHAMP is a catalogue of *variants*. It records whether an inhibitor has ever been reported for a variant, but carries no per-patient factor activity, antigen measurement, or cross-reacting material (CRM) typing. If the ceiling reflects missing patient-level information rather than insufficient model capacity, a resource supplying that information should move it.

A. Contributions

1. An inhibitor-risk model built at the *patient level* on the EAHAD/HADB resource — one observation per allele report rather than per variant — reaching AUC 0.786 ± 0.004 under strict variant-grouped validation. We are not aware of prior patient-level modelling on this resource but make no priority claim, as no systematic review was conducted.
2. Evidence that the genotype-only ceiling of approximately 0.74 replicates on an independently curated registry, establishing it as a property of the data class rather than an artefact of one dataset or pipeline.
3. Quantification of three leakage mechanisms — outcome-derived features, relabelling of unrecorded outcomes, and ungrouped partitioning — with the AUC attributable to each.
4. Identification and quantification of 64.5% variant overlap between CHAMP and EAHAD, which invalidates naive

cross-registry validation in this domain, together with a corrected external estimate.

5. Two experiments whose outcomes contradicted our hypotheses, reported with the claims revised rather than the framing.
6. A bounded calibration procedure preventing the deployed model from emitting risk estimates the underlying cohort cannot support.

II. RELATED WORK AND THE REPRODUCIBILITY PROBLEM

Published analyses of *F8* registries share a common design: rows are variants, features are registry columns (often label-encoded directly), the outcome is a reported inhibitor history, and evaluation is a stratified split or *k*-fold cross-validation. Three properties of this design account for the reported performance.

Identifier features. Columns such as the HGVS cDNA description are near-unique per row. Label-encoded, they function as primary keys. In our prior work a random forest given such a column attained training AUC 1.000 *with the outcome labels randomly permuted* — the diagnostic signature of an identifier rather than a predictor.

Relabelling of unrecorded outcomes. Registry inhibitor fields are tri-state: reported positive, reported negative, and not reported. Mapping "not reported" to negative converts absence of ascertainment into evidence of absence. In CHAMP this affects 1,744 of 4,040 rows and reduces apparent prevalence from 20.1% to 11.4%, raising the majority-class baseline against which accuracy is measured.

Resampling before partitioning. Random over-sampling duplicates minority rows verbatim. Applied before the train/test split, it places byte-identical copies of training rows into the test set. The reported 97.37% accuracy [1] is obtained under this protocol.

These are not exotic failures. They are the default outcome of applying a standard tabular machine-learning recipe to a curated biological registry. Our contribution is to measure each rather than to assert it.

III. DATA

A. The EAHAD/HADB Cohort

The resource [13] is distributed as two supplementary tables at different levels of analysis: a variant table (6,211 rows: mutation class, protein consequence, FVIII domain, exon, codon and nucleotide change) and an allele-report table (10,064 rows) carrying individual baseline FVIII activity, clinical severity, antigen, CRM type, reporting centre and source publication alongside the inhibitor outcome. The latter is the layer absent from CHAMP, and it determines the design of this study: the unit of analysis is the patient, with variant annotation joined on the variant identifier.

B. Outcome Definition

The inhibitor field takes eight surface forms normalising to three states: 4,130 negative, 836 positive, and 5,098 unrecorded. Unrecorded outcomes were retained as unlabelled and never mapped to negative. The analytic cohort is therefore 4,966 records at 16.83% prevalence, which falls within the published range for haemophilia A without adjustment — itself evidence that the tri-state field is being read correctly.

C. Cohort Characteristics

TABLE I

Baseline characteristics of the 4,966 labelled allele reports (836 inhibitor-positive, 16.8%), across 2,643 variants and 342 reporting studies. Median records per variant 1, maximum 104.

Characteristic	<i>n</i>	%	<i>Inh. rate</i>
<i>Mutation class</i>			
Missense	2,677	53.9	8.3%
Frameshift	1,072	21.6	22.4%
Nonsense	594	12.0	30.1%
Splice	250	5.0	17.2%
Large deletion	237	4.8	51.9%
In-frame	97	2.0	27.8%
Silent	30	0.6	3.3%
<i>Clinical severity</i>			
Severe	2,718	54.7	24.1%
Moderate	774	15.6	8.9%
Mild	1,392	28.0	6.2%
Unknown	82	1.7	29.3%
<i>Baseline FVIII activity (IU/dL)</i>			
< 1	2,356	47.4	20.5%
1–2	154	3.1	8.4%
2–5	610	12.3	7.4%
5–15	555	11.2	7.6%
15–40	415	8.4	4.1%
> 40	30	0.6	0.0%
Not measured	846	17.0	—

Median measured activity was 0.50 IU/dL (IQR 0.50–5.00); 1.00 among inhibitor-negative and 0.50 among inhibitor-positive records. FVIII antigen was measured in only 649 records (13.1%) and CRM type recorded in 1,312 (26.4%). The monotone gradient across mutation classes and across severity strata reproduces established immunology and constitutes a prior check on label quality.

D. The CHAMP Comparison Cohort

For external validation we used the CDC CHAMP *F8* variant list [2]: 4,040 variants, of which 2,296 carry a recorded outcome at 20.1% prevalence. CHAMP is variant-level and carries reported severity but no per-patient measurements.

E. Audit of Redistributed Derived Files

Two convenience files circulate alongside the raw supplement, prepared by merging and aggregating the two tables. Both reproduce, on this new dataset, the failure modes of Section II. We audit them because they are the form in which many investigators will encounter this resource.

Relabelling. The machine-learning-ready file contains 3,706 variant rows, of which 1,063 carry a negative label although no inhibitor outcome was ever recorded for that variant. Apparent prevalence falls from 23.5% (among variants actually followed up) to 13.3%.

Outcome-derived features. The merged file exposes twelve columns in our forbidden set, including a per-variant inhibitor positive rate — the outcome itself, aggregated over the very patient rows being predicted. Adding that single column to our

design matrix moves AUC from 0.777 to 0.966, a change of +0.189, with no biological information added.

IV. METHODS

A. Feature Construction

Ninety-five numeric descriptors were derived in six declared blocks: genotype class (21), FVIII domain (16), protein position (13), residue chemistry (18) using Grantham distance [3] and BLOSUM62 [4], patient clinical measurements (18), and reporting context (9). Every feature is either a property of the variant or a measurement recorded at diagnosis; none is derived from the outcome, and none is row-unique.

Two engineering decisions required judgement. The curated CpG column is empty in this release, so the hotspot signature was derived from the nucleotide change, since C>T and G>A transitions are the deamination products of a methylated CpG dinucleotide. The factor-level field mixes plain numerals, left-censored readings, ranges and annotated entries; censored values were assigned half the stated bound, the conventional substitution for a left-censored assay reading, which preserves the ordering of a censored value below the corresponding observed one. Missingness is encoded explicitly through indicator features, so median imputation does not silently create a measurement indistinguishable from an observed one.

B. Validation Protocol

2,643 variants generate 4,966 records, with recurrent variants contributing up to 104 records each. Under random partitioning a model can memorise a variant during training and be scored on that same variant at test — the patient-level form of the identifier leak described in Section II. Every partition in this study is therefore grouped by variant.

As a stricter test of transportability we repeat validation grouping by source publication. Reporting centres differ in inhibitor screening practice and in which patients they publish, so study-grouped performance estimates behaviour in a centre not represented in training.

Class imbalance is addressed with class weighting exclusively. No over-sampling, under-sampling or synthetic minority generation is applied at any stage.

Because a single 20% held-out partition contains approximately 159 positive records and carries roughly ± 0.04 of sampling noise, the primary estimate is five-fold variant-grouped cross-validation repeated over three seeds on the entire labelled cohort. A single grouped held-out partition, scored once after model and thresholds were fixed, serves as an independent check.

C. Models and Ensembling

Twelve families were screened, spanning linear, kernel, neighbour, forest, boosting and neural approaches, including XGBoost [5] and LightGBM [6]. Leading families were tuned by randomised search over 25 draws scored on precomputed grouped folds.

The `scikit-learn` stacking meta-estimator cannot be used in this setting: it generates internal cross-validation folds at random with no mechanism to propagate grouping, placing the same variant on both sides of the meta-learner's training boundary. In our screen this produced an inverted model (AUC 0.241). We therefore construct the out-of-fold matrix

manually from grouped folds and compare probability averaging, rank averaging, logistic stacking, and greedy forward selection with replacement [7].

Members are combined by rank average rather than probability average, because members are calibrated on different scales. Ranks are computed against the training score distribution stored at fit time rather than within the scoring batch, so an individual patient's estimate is independent of which other patients are scored alongside them.

D. Calibration

Probabilities are calibrated by isotonic regression [8] fitted on training out-of-fold scores only. Unmodified isotonic regression returned exactly 1.0 for the highest-scoring inputs, because its terminal step covered a small, entirely positive training bin. A deployed tool reporting *certainty* of inhibitor development is not supportable: the highest-risk observed stratum develops inhibitors at 53.1%, and the highest calibration decile at 59.6%.

We therefore constrain the calibrator. Training scores are partitioned into deciles, each decile's positive rate is Laplace-smoothed as $(k+1)/(n+2)$, and predictions are clipped to the interval spanned by those smoothed rates — [0.025, 0.520] in this cohort. The transformation is monotone, so ranking and hence AUC are unchanged; only the reported magnitude is constrained to what the cohort supports.

E. Statistical Analysis

Discrimination is reported as AUC-ROC and AUC-PR against the prevalence baseline. Uncertainty is estimated by stratified bootstrap over 2,000 resamples. Paired model comparison uses DeLong's test [9]. Calibration is summarised by the Brier score and expected calibration error. Clinical utility is assessed by decision-curve net benefit [10]. Operating thresholds are selected on training folds by Youden's J [11], by accuracy maximisation, and at fixed sensitivity targets.

V. RESULTS

A. Model Screen

TABLE II

Twelve model families under five-fold variant-grouped cross-validation on the training partition ($n = 3,992$). The final row documents a methodological failure and is not a candidate model.

Model	AUC-ROC	AUC-PR	Bal. acc.	Brier
Extremely rand. trees	0.7778	0.4568	0.7183	0.1514
Random forest	0.7774	0.4615	0.7176	0.1363
LightGBM	0.7745	0.4773	0.7065	0.1420
XGBoost	0.7735	0.4678	0.7110	0.1559
Gradient boosting	0.7705	0.4769	0.7120	0.1165
Hist. grad. boosting	0.7682	0.4592	0.7010	0.1479
Multilayer perceptron	0.7622	0.4361	0.7105	0.1202
Elastic-net logistic	0.7572	0.4409	0.7002	0.1883
L2 logistic	0.7552	0.4372	0.6995	0.1887
SVM (RBF)	0.7551	0.3997	0.7018	0.1214
k -nearest neighbours	0.7507	0.4078	0.6907	0.1244
Stacking (ungrouped CV)	0.2412	0.1108	0.5019	0.3492

The eleven valid families span 0.027 AUC. Model capacity is not the binding constraint on this problem — a finding that

recurs throughout. Tuning improved the leading families to 0.781–0.784. Greedy forward selection retained three members (XGBoost, extremely randomised trees, LightGBM) at out-of-fold AUC 0.7859; rank and mean averaging over all five tuned members gave 0.7860, and the three-member ensemble was retained for inference economy.

B. Feature Ablation

TABLE III

Contribution of each feature block under repeated variant-grouped cross-validation over three seeds.

<i>Rung</i>	<i>k</i>	<i>AUC-ROC</i>	<i>A</i>
Genotype class only	21	0.7024 ± .0013	—
+ FVIII domain	37	0.7261 ± .0033	+0.0237
+ protein position	50	0.7372 ± .0012	+0.0111
+ chemistry (all genomic)	68	0.7417 ± .0029	+0.0045
+ patient phenotype	86	0.7645 ± .0011	+0.0228
+ reporting region	95	0.7779 ± .0025	+0.0134

Two findings follow. First, the genomic rung reaches 0.7417. This reproduces the ceiling previously established on CHAMP — a registry curated by a different consortium, from different patients, through an independently written feature pipeline. Two registries agreeing on where genotype-only information is exhausted is substantially stronger evidence than either alone, and reframes the ceiling from a limitation of one analysis into a property of this class of data. Second, patient baseline phenotype contributes +0.023 AUC, confirming the hypothesis that motivated this study.

C. Leakage Controls

TABLE IV

Controls bounding the primary result.

<i>Control</i>	<i>Result</i>
Permuted labels (prior work: 1.000)	0.5042
Ungrouped random CV	0.8098
Variant-grouped CV	0.7779
Study-grouped CV	0.7797 ± .0004
Test rows identical to a training row	1.03%
Strongest feature–outcome $ r $	0.247

The 0.032 AUC forgone by grouping is of the same magnitude as the entire clinical-layer gain of 0.023, illustrating how readily a partitioning choice can be mistaken for a modelling advance. Random over-sampling before partitioning leaves roughly half the test set byte-identical to training rows; the observed 1.03% reflects ordinary coincidence between patients who share a mutation class, a severity band and the same missing measurements.

D. Primary Result and Operating Characteristics

Under repeated variant-grouped cross-validation over the full labelled cohort ($n = 4,966$), the ensemble achieved **AUC-ROC 0.7861 ± 0.0040** (per-seed 0.7870, 0.7808, 0.7905).

On the held-out partition ($n = 974$, prevalence 16.3%): AUC-ROC 0.7451 (95% CI 0.6927–0.7890); AUC-PR 0.4552 (0.3883–0.5295) against a no-skill baseline of 0.1632; Brier 0.114; expected calibration error 0.033. The cross-validated estimate lies within the held-out confidence interval.

TABLE V

Operating characteristics on the held-out partition ($n = 974$, 159 positive). Majority-class baseline accuracy is 83.68%. Thresholds were selected on training folds. The balanced and rule-out points are those at which the tool is specified; the accuracy-maximising point is reported for comparison only.

<i>Operating point</i>	<i>Threshold</i>	<i>Accuracy</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>PPV</i>	<i>NPV</i>	<i>MCC</i>	<i>TP</i>	<i>FP</i>	<i>FN</i>	<i>TN</i>
Accuracy-maximising	0.474	86.65%	25.8%	98.5%	77.4%	87.2%	0.396	41	12	118	803
Balanced (Youden's J)	0.181	76.69%	61.6%	79.6%	37.1%	91.4%	0.343	98	166	61	649
Rule-out (80% sens. target)	0.131	65.61%	74.2%	63.9%	28.6%	92.7%	0.285	118	294	41	521
Rule-out (90% sens. target)	0.083	49.38%	79.9%	43.4%	21.6%	91.7%	0.176	127	461	32	354

Accuracy is reported only alongside its baseline. The accuracy-maximising threshold exceeds the no-skill baseline by 2.97 percentage points while identifying 25.8% of inhibitor-positive patients; it is the least clinically useful row in Table V, and the one that an evaluation rubric specifying "high accuracy" would select. The tool is specified at the balanced and rule-out points, where negative predictive value exceeds 0.91.

Decision-curve net benefit was 0.088 at a 10% threshold probability (treat-all 0.070) and 0.058 at 20% (treat-all –0.046). The model exceeded both treat-all and treat-none strategies for threshold probabilities between 0.09 and 0.52, covering 73% of the swept range. Below a 9% threshold, monitoring every patient is preferable and the model offers no advantage.

Against a genomic-only model on the same held-out patients (AUC 0.7305), the full model's advantage of 0.0147 did not reach significance (DeLong $z = 1.06$, $p = 0.29$), which is expected at this sample size. The ablation in Table III, estimated over the full cohort with repeated resampling, is the better-powered comparison.

E. Cross-Registry Validation and a Registry-Overlap Artefact

Projecting both registries into a 57-feature intersection, a model trained on HADB and applied to CHAMP without refitting scored AUC 0.879 — exceeding CHAMP's own within-registry cross-validation of 0.725.

A model applied to an unseen cohort should not outperform that cohort's own cross-validation. We treated this as a diagnostic rather than a result. CHAMP and EAHAD are both curated from the published literature, and they curate substantially the same publications. Matching on mutation class, mature residue, reference residue and substituted residue, 1,259 of 1,951 (64.5%) of CHAMP's substitution-resolvable labelled variants are already present in HADB.

One methodological subtlety governs this comparison. The matching key requires a reference residue and position, which frameshift and large structural variants do not provide. Left uncorrected, every frameshift is classified as novel by default, and the resulting comparison contrasts a 77%-frameshift subset with a 60%-missense subset, measuring mutation-class

composition rather than novelty. Both strata are therefore restricted to substitution-like variants.

TABLE VI

Transfer from HADB to CHAMP, stratified by prior presence of the variant in the training registry.

Stratum	<i>n</i>	Prev.	AUC-ROC
Present in HADB	1,259	0.137	0.9361
Novel to HADB	692	0.240	0.8513
Attributable contamination	—	—	0.0848

The corrected external estimate is AUC 0.8513 (95% CI 0.8193–0.8807). On the identical 692 novel records, CHAMP's own cross-validated model achieved AUC 0.6237. Training on 4,966 patient-level European records therefore predicted novel North American variants 0.2276 AUC better than CHAMP predicted them from its own data — independent confirmation, by a different route, of the ablation's conclusion that repeated patient-level observation constitutes richer supervision than a variant catalogue.

F. Reproduction of the Relabelling Effect

TABLE VII

Identical fitted model and variant set; only the label convention differs.

Label convention	<i>n</i>	Prev.	Base	Acc.	AUC
Recorded outcomes only	2,296	0.201	79.9%	80.1%	0.879
Unrecorded → negative	4,040	0.114	88.6%	74.7%	0.855

Discrimination does not improve under relabelling; AUC in fact declines. What changes is the accuracy obtainable *without skill*, which rises from 79.9% to 88.6%. A headline accuracy near 97% is measured on a set whose no-skill baseline is already 88.6%, and is approached by tuning the decision threshold toward the enlarged majority class.

G. Results Contrary to Hypothesis

We predicted that collapsing records to one row per variant would lose information. It scored *higher*: 0.8038 against 0.7766. The two estimates are not comparable, differing in unit of analysis, label definition and sample size, and predicting a variant's *modal* outcome is an easier problem than predicting an individual's, since feature averaging across a variant's records cancels measurement noise and well-characterised recurrent variants dominate the row set.

The case for patient-level modelling therefore rests on a different measurement. Among the 537 variants with two or more recorded outcomes, 124 (23.1%) are discordant — patients carrying an identical variant who differ in inhibitor outcome — spanning 1,297 records. Aggregation assigns every such patient the same prediction and is necessarily wrong for the minority, and it cannot use per-patient factor activity at all. The second contrary result, naive cross-registry transfer overstating external validity, is detailed in Section V-E.

H. Feature Attribution

Permutation importance on the held-out partition ranked severity ordinal (0.0084 AUC), East Asian reporting region (0.0073), nucleotide position (0.0046), log FVIII activity (0.0044) and multi-domain involvement (0.0043) highest.

Aggregated by block: clinical 0.0287, position 0.0211, context 0.0148, chemistry 0.0116, domain 0.0088, genotype 0.0042.

VI. DISCUSSION

A. What Limits Performance

The evidence converges on a single conclusion: the constraint is informational, not architectural. Eleven model families span 0.027 AUC; tuning contributed 0.005 and ensembling 0.003. By comparison, the partitioning leak we declined to exploit was worth 0.032, and a single outcome-derived column was worth 0.189.

The genomic ceiling of approximately 0.742, now observed in two independently curated registries, is best interpreted as the information content of *F8* genotype with respect to inhibitor development. Established determinants that no public registry records — treatment intensity, exposure days at first bleed, product class (plasma-derived versus recombinant [12]), surgery or infection at first exposure, and HLA class II typing — are absent by construction. Their absence, not model capacity, bounds these estimates.

B. Direction of Effect for Clinical Features

Baseline factor activity could in principle be a *consequence* of the outcome rather than a predictor of it, since a circulating inhibitor suppresses measured FVIII activity. The registry documents these as diagnostic baseline values but does not timestamp them relative to inhibitor detection, so this concern cannot be fully resolved with these data.

We therefore report the genomic-only rung alongside every headline. A reader who rejects the clinical block entirely retains a defensible 0.742 model validated across two registries. The clinical contribution is presented as an improvement conditional on that assumption, not as a foundation.

C. The Reporting-Region Block

The context block contributes +0.0134 AUC, and East Asian reporting region ranks second in permutation importance. This warrants explicit caution. The registry records the reporting laboratory's country, not patient ancestry. Observed regional inhibitor rates vary widely (East Asia 45.5%, *n* = 308; South Asia 10.8%, *n* = 297; North America 90.9%, *n* = 11), and the extreme values rest on very small strata.

Ancestry is a genuine, established risk modifier for inhibitor development, so part of this signal may be biological. But it is confounded with centre-level screening intensity and publication selection, and we cannot separate the two here. A deployment unwilling to accept that confound can use the model without the context block at AUC 0.7645, and we recommend this for any use outside the regions represented in training.

D. An Anomaly in CRM Typing

CRM type did not behave as the underlying immunology predicts. CRM-negative (type I) patients, those with no circulating antigen, showed a *lower* inhibitor rate (3.8%, *n* = 156) than CRM-unclassified patients (22.6%, *n* = 953). We attribute this to ascertainment: CRM typing is performed selectively, largely on mild and moderate patients with discrepant one-stage and chromogenic assays, who are at low baseline inhibitor risk. This is a selection effect in the registry

rather than a contradiction of the immunology, but it means CRM type should not be interpreted causally in this cohort.

E. Implications for Cross-Registry Validation

The 64.5% overlap between CHAMP and EAHAD has consequences beyond this study. Cross-registry transfer is widely regarded as strong evidence of external validity, and in this domain a naive application of it inflated our own estimate by 0.085 AUC. Investigators validating *F8* models — and plausibly models built on other curated disease-variant resources compiled from shared literature — should establish variant-level disjointness before reporting external performance, and should report the matching procedure, since key resolvability interacts with mutation class in a way that can silently confound the comparison.

F. Clinical Positioning

The model is a research tool and is not a validated medical device. It does not replace laboratory inhibitor testing or clinical judgement. Its intended use is prioritisation of monitoring intensity during the first fifty exposure days. At the balanced operating point it identifies 61.6% of patients who go on to develop inhibitors, at a cost of 166 false positives per 974 patients, with negative predictive value 0.914. The clinical consequence of a false positive is intensified monitoring, not a change in diagnosis or a withheld therapy — the asymmetry that makes a rule-out threshold defensible at this level of discrimination.

VII. LIMITATIONS

1. Both cohorts are compiled from published reports rather than consecutive series. Patients with inhibitors are plausibly more likely to be published, and screening intensity varies by centre.
2. The temporality of baseline factor measurements relative to inhibitor detection is unresolved (Section VI-B).
3. Causal covariates — treatment intensity, exposure days, product class, HLA typing — are absent (Section VI-A).
4. Reporting region is a confounded proxy for ancestry (Section VI-C).
5. The held-out partition is small ($n = 974$, 159 positive), giving wide intervals; the repeated cross-validated estimate is the primary result for this reason.
6. Scope is a single gene and a single disease; transfer to haemophilia B (*F9*) was not attempted.
7. All estimates are retrospective and registry-based. No prospective validation was performed.

VIII. CONCLUSION

Genotype-only prediction of FVIII inhibitor risk saturates near AUC 0.74, a value we observe independently in two separately curated registries and which is therefore best understood as a property of the available data rather than of model capacity. Patient-level records carrying individual baseline phenotype raise achievable discrimination to 0.786 ± 0.004 under strict variant-grouped validation, with calibrated probabilities and retained performance in unseen reporting centres.

Published accuracies above 95% do not represent predictive performance. Each contributing mechanism —

outcome-derived features, relabelling of unrecorded outcomes, and resampling before partitioning — is reproducible on demand and quantified here.

Progress beyond approximately 0.79 will not come from better classifiers. It requires prospective cohorts recording treatment intensity, exposure history, product class and HLA type. Until such data are available, the honest contribution of genotype-based modelling is a calibrated rule-out instrument, not a decisive diagnostic.

DATA AND CODE AVAILABILITY

All code, trained artefacts, result files and the executed analysis notebook are available at github.com/dchauthari7177/hemophilia. Every result is regenerated by the released scripts and serialised to JSON. The analysis notebook executes eight integrity assertions, including a direct test for the over-sampling signature, and fails if any does not hold. The test suite comprises 128 tests.

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